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The application of geospatial analyses to support an integrated study into the ecological character and sustainable use of Lake Chilwa

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ABSTRACT

Wetlands, which occupy a considerable proportion of Malawi, are an important resource for the rural-poor. In regions of low and erratic rainfall, they provide important opportunities for agriculture and fisheries and are often vital for food security. However, overutilization and some farming practices can result in environmental degradation and the loss of valuable ecosystem services. The Lake Chilwa wetland, a complex aquatic ecosystem comprising a shallow open water lake surrounded by marsh and floodplain grassland, is one of the most important wetlands in Malawi. Most of the 1 million people who live in the vicinity of the lake derive their livelihoods from the wetland. Increases in population in recent decades have increased the pressures on the wetland. In light of continued population growth, it is anticipated that some levels of resource use may be difficult to sustain in the long-term. Consequently, planning the use of wetland resources is a priority if the diverse benefits that local communities presently receive are to be sustained. Baseline wetland information is needed to determine appropriate sustainable management plans, as well as to identify limits to resilience. However, for many wetlands in southern and eastern Africa, these data are lacking, or not available to those who need them. This paper provides an overview of the distribution and characteristics of wetlands in Malawi. Using Lake Chilwa as an example, it illustrates how Earth Observation data, in combination with hydrometric and botanical data, can be used to provide information crucial for sustainable wetland management.

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Introduction

In Malawi, wetlands are of major importance for much of the rural population. By providing water for agricultural and domestic uses as well as many other natural resources, including fisheries, they sustain livelihoods and increase food security (Scoones, 1991; PEMconsult, 1997). Increasing human population is, however, escalating socio-economic pressure to extend wetland agriculture and fisheries, and there is increasing concern about the environmental degradation and loss of ecosystem services arising from increased utilization (Meadows, 1985; McFarlane and Whitlow, 1990; Schuyt, 2005). In many African wetlands agricultural practices, both within and around wetlands has modified the hydrology and caused pollution, siltation, and other forms of degradation, and in many instances undermined important fisheries (Thieme et al., 2005). Similarly, fisheries practices have adversely impacted fish stocks and, in places, have also affected the hydrology of wetlands (e.g., through blocking and unblocking channels) that, in turn, has had direct and indirect impacts on agriculture (e.g., Vershuren et al., 2002; Abebe and Geheb, 2003; Schuyt, 2005). In line with conclusions drawn from analyses elsewhere (Wood and van Helsma, 2008), the integration of agricultural and fisheries management practices is seen as a prerequisite for sustainable wetland development in many African wetlands.

Currently, the basis for making decisions about the extent and manner in which wetlands can be utilized for agriculture and fisheries is weak and exacerbated by lack of knowledge about their vulnerability and resilience to change. As with all natural systems, the resilience of wetland ecosystems derives from both a duplication of ecological functions across a range of spatial and temporal scales and a diversity of functions operating at each scale (Gunderson et al., 2002). By reducing functional diversity, agricultural and fishery practices can increase wetland vulnerability and reduce their ability to recover from disturbances. It has been widely shown that overexploitation of wetland resources can have profound social and economic repercussions for people dependent on the ecosystem services they provide (Revenga et al., 2000; Kura et al, 2004; MEA, 2005).

Previous analyses of the interactions between agriculture (including fisheries) and wetlands around the world have highlighted the social perspectives involved with wetland utilization and concluded

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that, although conditions are context-specific, economics issues are the predominant driver of change (Kura et al., 2004; Ratner et al., 2004; Wood and van Helsma, 2008). Although mediated by cultural and institutional factors, such as markets and policies, the dynamics of change are also increasingly being influenced by global markets (Wood and van Helsma, 2008). The same studies have noted that policy and management responses to wetland degradation have largely been unsuccessful, particularly in developing countries, primarily because of a lack of basic information on wetlands, little understanding of wetland processes and dynamics, and failure to perceive wetlands as complex, interlinked social and ecological systems.

In Malawi, if further erosion of the wetland resource base and associated social decline are to be avoided, basic information is needed to develop policies that safeguard wetland integrity and promote equitable, sustainable use. For this to occur, greater understanding of the relationships between wetland utilization, biophysical characteristics, and socioeconomic factors is essential. As a contribution to this task, this paper provides baseline information on land cover and use and the ecological character of Lake Chilwa, in particular the hydrology and vegetation patterns that support many of the ecosystem services that people derive from wetlands.

Wetlands of Malawi: an overview

Despite widespread distribution across Africa, knowledge of the wetland resource is far from complete and is generally inadequate to support management needs (Stevenson and Frazier, 1999; Thieme et al., 2005; Schuyt, 2005; Rebelo et al., 2009). As is the case for many African countries, Malawi lacks a detailed wetland inventory. Given the absence of a national inventory, the Global Lakes and Wetlands Database (Lehner and Döll, 2004) has been used to provide an overview of wetland habitats in Malawi (Fig. 1a). While global data sets provide a useful source of information on the extent and distribution of wetland habitats at a broad scale, estimates of wetland extent from such sources are likely to be underestimates (Finlayson et al., 1999; Rebelo et al., 2009). According to the information from this data set, freshwater marshes and floodplains in Malawi cover an area of 5360 km^2 and lakes, 23,506 km^2. In this data set, "lakes" generally refer to permanent bodies (lentic water bodies) without direct connection to the sea. In addition, to account for the mixed classifications used in the source data sets, saline lakes and lagoons (but not 'lagoon areas') are also included as lakes, while intermittent or ephemeral water bodies are excluded. The definition of the wetland classes (including "freshwater marshes and floodplains" has generally followed that of the Ramsar Convention; RIS, 2009). For more information on the classification terms, the reader is directed to Lehner and Döll (2004).

The distribution of marshes (herbaceous, frequently flooded wetland) and floodplains and lakes across the terrestrial ecoregions of Malawi, as outlined by Olson et al. (2001), is shown in Fig. 1b. Excluding Lake Malawi, the largest area of lakes (789 km^2) occurs within the flooded grassland ecoregion (Table 1), while 91% of Malawi's freshwater marsh and floodplain habitats are found within the tropical and subtropical grassland ecoregion. The freshwater marsh and floodplain habitats of Malawi can be further separated into four main classes: riverine and lacustrine floodplains which are typically dominated by grasses; dambo wetlands with grasses in permanently waterlogged (but not deeply flooded) or temporarily flooded sites; permanent swamps and marshes which are dominated by *Phragmites* and *Typha* or *Cyperus* species in permanent water with seasonally fluctuating levels; and sedge-marshes which occur where inundation is neither very deep nor permanent, at sites where the surface dries for a short period each year (Hughes et al., 1992). This latter type is often found on the landward side of permanent swamps. Malawi has nominated one site, Lake Chilwa, as a wetland of international importance (known as a Ramsar site) under the Ramsar Convention on Wetlands. Located within the flooded grassland ecoregion in the southern part of the country, it is this wetland on which this paper focuses.

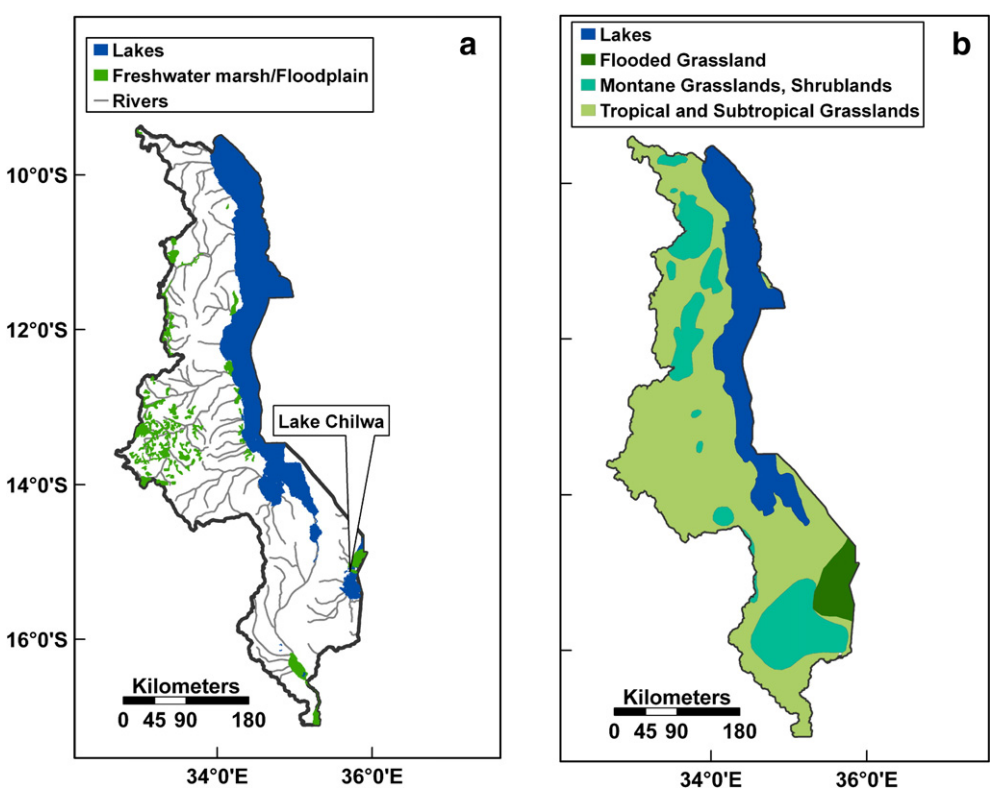

Fig. 1. Distribution of (a) lakes, rivers, wetlands (Lehner and Döll, 2004) and (b) ecoregions (Olson et al., 2001) of Malawi.

Table 1
Presence of wetland habitats within ecoregions of Malawi.

Wetland type (GLWD: Lehner and Döll, 2004)	Wetland area (km^2)	% of Malawi	WWF ecoregion (Olson et al., 2001)
Lakes	789	0.70	Flooded grassland
	18	0.02	Montane grasslands, shrublands
	1150	0.97	Tropical and subtropical grasslands
	21,550	18.2	Lakes
Freshwater marsh/ floodplain	396	0.33	Flooded grassland
	103	0.09	Montane grasslands, shrublands
	4851	4.09	Tropical and subtropical grasslands

Study area: the Lake Chilwa ecosystem

Located to the southeast of Lake Malawi (Fig. 1), Lake Chilwa with an area of approximately 2284 km^2 is the second largest lake in the country. Situated in a shallow basin, the lake lies below 623 m above sea level (masl; Fig. 2) on the border with Mozambique, it is considered to be one of the most productive lakes in Africa (Schuyt, 2005). Located in a tectonic depression, the lake has a radial drainage pattern with 14 streams converging from the surrounding watershed. Mt. Zomba and lower hills are located to the west, the Mlanje massif to the south, lower hills in Mozambique to the east and a sandbar, which divides Lake Chilwa from Lake Chiuta in the north. More information on the geology of the area is provided in Morgan and Kalk (1970).

The Lake Chilwa ecosystem consists of a shallow, enclosed endorheic saline lake with an average depth of 1–2 m (RIS, 2009). It is surrounded by an area of marshland dominated by dense stands of *Typha* (in many places known as reed mace or cattail), with a seasonally inundated grassland floodplain forming a belt around these. The lake and surrounding marsh and floodplain were designated as a Ramsar site in 1996 under Criterion 6, i.e., for regularly supporting 1% of the individuals in a population of one species or subspecies of water bird. With approximately 162 people per km^2 the Lake Chilwa catchment has one of the highest population densities in Malawi (Bhima, 2006). As a result, one of the biggest threats to the wetland is increased exploitation of resources within the wetland itself as well as within the surrounding catchment due to increased population pressure, resulting in serious degradation and loss of biodiversity and interconnected ecosystem services.

The hydrology of the lake is an important control on the ecology of the wetland, influencing not only the water chemistry and physical properties but also the composition of the vegetation and soil characteristics (Howard-Williams and Walker, 1974; Beadle, 1981). Based on the catchment boundary derived from elevation data extracted from the Shuttle Radar Topography Mission (SRTM) data set, the catchment is 8784 km^2 of which 5724 km^2 (65%) is in Malawi and 3060 km^2 (35%) is in Mozambique. These figures differ slightly from those given in the current Lake Chilwa wetland management plan (EAD, 2001) but are based on the best digital elevation data currently available at the catchment level. Average annual rainfall across the catchment is approximately 1362 mm. Water levels fluctuate between 621.3 and 625.5 masl with peak water level typically occurring at the end of the rainy season, in April. Seasonal variations are superimposed on longer wetter and drying cycles (Fig. 3). Occasionally the lake dries completely. In the last century alone, it has dried and filled eight times, most recently in 1996/1997.

Materials and methods

Hydrology

Several rivers flow into Lake Chilwa, but only the six most prominent rivers (i.e., the Sombani, Namdzi, Likangala, Thondwe, Domasi, and Phalombe) are perennial. Abstractions for human use are relatively low, about 0.15 Mm^3 per day, with by far the largest abstractions (0.12 Mm^3 per day) being for irrigation. However, there is concern that, as population in the catchment rises, abstractions will increase (EAD, 2001). At present, there is no water management plan for the catchment. Water level is monitored at gauge boards located on four of these rivers and converted to flows using rating equations (Table 2). Computed flows are aggregated to provide estimates of monthly flow volumes. The monthly flow data were obtained from the Water Resources Department of the Ministry of Irrigation and Water Development.

Analyses of these time series highlighted both the high seasonal variation in flow (i.e., typically 77–85% of total annual runoff occurs

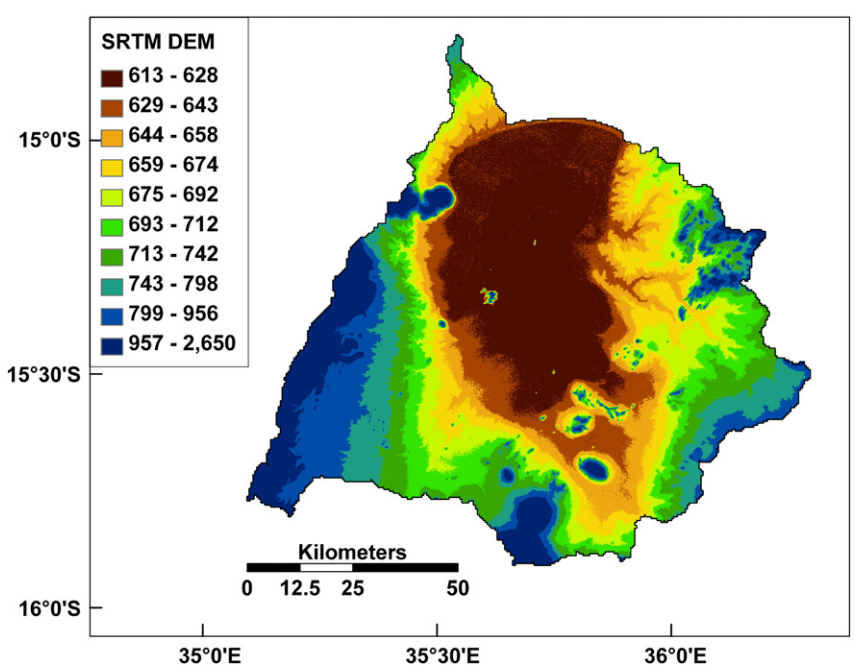

Fig. 2. Elevation derived from Shuttle Radar Topography Mission (SRTM) data for the Lake Chilwa catchment.

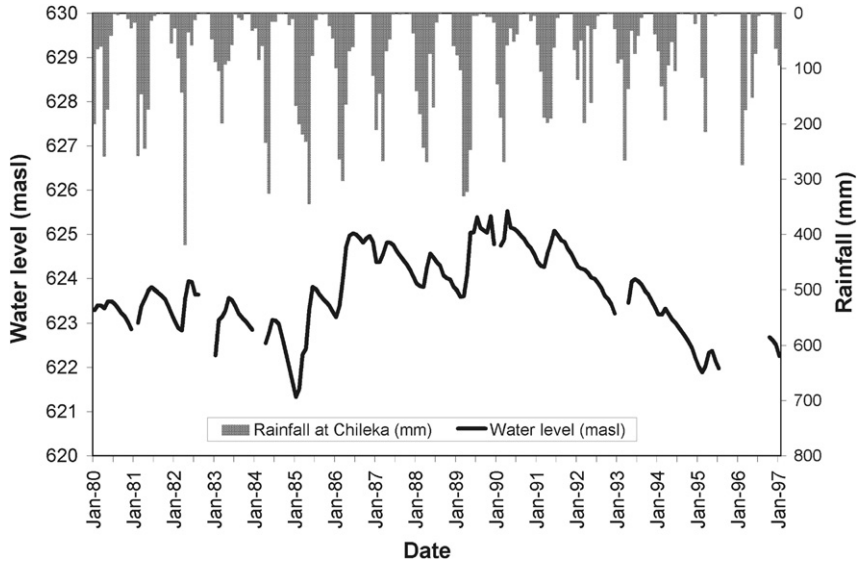

Fig. 3. Water levels measured in Lake Chilwa from January 1984 to December 1994 (some values in-filled by interpolation) and rainfall measured at the Chileka rain gauge (the nearest gauge, located at approximately the same altitude as the lake (i.e., 767 masl) and with a reliable record) over the same period (sources of data: Ministry of Irrigation and Water Development, Malawi, and FAO Loc-Cim database).

between January and May) and the considerable interannual variability in flow in all the rivers. Although a lack of data means many assumptions are necessary, a crude estimate of the wetland water budget was derived. The water budget of the wetland was approximated by the equation:

$$P + Q_i = E + Q_o, \quad (1)$$

where

P	precipitation onto the wetland
Q_i	total inflow into the wetland
E	evapotranspiration from the wetland (a combination of open water evaporation and evapotranspiration from the marsh and grass)
Q_o	outflow

To the authors' knowledge, there have been no studies of either regional or local groundwater movement in the vicinity of the lake. Since it is located in a tectonic depression, it is likely that lateral subsurface flows move towards the lake, and it is possible that groundwater comprises a large proportion of the total inflow into the lake, particularly in the dry season. However, more research, perhaps using natural isotope tracers, is required to determine whether or not this is the case.

Mean annual evapotranspiration from the wetland was estimated based on different wetland vegetation types (EAD, 2001) and estimates of likely wet and dry season evapotranspiration losses. Due to lack of observed data, a number of assumptions were necessary, including the following:

- There is no outflow from the lake as either surface water or groundwater. There is no surface water outlet from the lake and, although it cannot be validated without further study, it was also assumed that groundwater outflows are negligible.
- Average annual rainfall directly into the wetland is 1000 mm, with 80% falling in the wet season (i.e., as indicated by the Chileka rainfall records).
- Average annual potential evapotranspiration from the floodplain grasses is 1400 mm (i.e., estimated potential evapotranspiration derived for a short well-watered grass using the FAO Penman–Monteith formula (Penman, 1948; Allen et al., 1998)). Mean monthly estimates of potential evapotranspiration were obtained from the FAO Loc-Clim database (http://www.fao.org/sd/locclim/srv/locclim.home). Actual evapotranspiration from the floodplain grasses was assumed to be at potential rates in the wet season but was slightly reduced in the dry season (to 1200 mm) to allow for water deficits (Gasca-Tucker and Acreman, 2000).
- Open water evaporation is always at potential rates and is greater than that from grass (Allen et al., 1998; Grismer et al., 2002). This was derived by applying a correction factor of 1.3 to the FAO Penman–Monteith estimate derived for grass to give an annual value of 1800 mm.
- Actual evapotranspiration from the *Typha* marshes was greater than grass because the higher aerodynamic roughness of the taller reeds induces turbulence above them and hence enhances evapotranspiration (Herbst and Kappen, 1999; Acreman et al., 2003). However, it remains slightly less than open water potential and annually was assumed to be 1700 mm. Water stress was assumed to be minimal so the same value was used for both the wet and dry season.
- Annual actual evapotranspiration from cultivated areas was assumed to be similar to the FAO Penman–Monteith potential for grass but was fixed at 1350 mm because these areas are irrigated and so water deficits are low.

Based on these assumptions, total inflow to the lake (i.e., comprising both surface flow and any groundwater inflow) was estimated from Eq. (1).

Table 2
Gauging stations for which monthly flow data are available.

Gauge number	River	Length of record	Nos. of complete years	Latitude	Longitude	Catchment area (km^2)
20221	Likangali at Nkokanguqo	1959–2002	20	15.41°S	35.44°E	144.0
20102	Sombani at Phaloni Hill	1951–1979	24	15.63°S	35.76°E	843.0
20303	Domasi at Domasi	1957–1998	24	15.28°S	35.38°E	72.8
20222	Thondwe at Jali	1959–2002	19	15.49°S	35.48°E	302.0

Vegetation mapping

A multispatial and multitemporal data set consisting of optical and radar remote sensing images (Table 3), aerial photos, and field data have been used to provide baseline information on the spatial and seasonal distribution of water and vegetation within the Lake Chilwa wetland. The data set consists of a Landsat 5 Thematic Mapper (TM) scene acquired on November 18, 2005, a scene acquired by the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) sensor on May 21, 2006, a radar scene acquired on May 17, 2006, by the Phased Array L-band Synthetic Aperture Radar (PALSAR) instrument in Fine Beam Dual mode, and two PALSAR images acquired on October 19, 2006, and the February 12, 2007, in Fine Beam Single mode. The aerial photographs were acquired in December 2005 by the Danish Hunters Association. Rainfall in the catchment falls between November and April, followed by a dry season from May to October. Full details on the data sets used and methodologies applied are given elsewhere (Rebelo, 2009).

Field surveys were conducted in October 2006 at 90 sites where a 20×20-m quadrat was used to record the vegetation species, hydrology, and land use. The location of the sample sites in shown in Fig. 4. A total of 90 sites that were accessible by land or boat were visited. At each site, the dominant plant species were recorded based on visual estimates of the above-ground biomass and ground cover. The information on the dominant plants was used to characterize the vegetation patterns determined from the remotely sensed imagery. Further information on the vegetation was obtained from the aerial photographs that were acquired at a sufficiently low altitude to visually identify the dominant vegetation type.

Analysis of the data involved classification of the remote sensing images into dominant wetland types. This was achieved through the use of a Decision Tree (DT) classifier, based on a series of binary decision rules. DT classifiers allow multistage classifications to be performed, recursively partitioning the input data set into increasingly homogenous subsets. A particular advantage of this approach is that data sets with different spatial resolutions, as well as ancillary data sets can be used together during the classification process. Image segmentation was performed at each node based on histogram analysis in order to separate the data into two mutually exclusive classes. The vegetation within the broader "wetland vegetation" class (i.e., excluding terrestrial land) was subsequently classified into species dominance based on a subset of the field sample sites (50% of the sites were randomly selected) and the aerial photograph locations. An accuracy assessment of the classification results based on an independent sample (the 45 field sites not used in the training phase) indicated a classification accuracy of 89%.

Analysis of fire activity

Fire products from the MODerate Resolution Imaging Spectrometer (MODIS) sensor provide a global time series of daily fire activity at a spatial resolution of 1 km. Fires are identified based on their absolute detection if the fire produces enough heat, or their detection relative to the thermal emission of the surrounding pixels for smaller and/or cooler fires (Justice et al., 2002). Twice-daily active fire observations from the MODIS sensors on board the Terra and Aqua satellites were acquired as 8-day composites, for the 12-month period January 1 to December 31, 2006. The Terra satellite has a morning overpass with an equatorial crossing time of 10.30 A.M., and the Aqua satellite an afternoon overpass with equatorial crossing of 1.30 P.M. Fires detected within the Lake Chilwa catchment at the time of the satellite overpass have been extracted from these data sets and processed to determine the afternoon and morning fire activity in each month when there was burning.

Table 3
Remote sensing data sets acquired over Lake Chilwa.

Sensor	Wavebands/mode	Pixel spacing (m)	Acquisition date
Landsat TM	Visible, near infrared	30	November 18, 2005
	Thermal	120	
ASTER	Visible, near infrared	15	May 21, 2006
	Shortwave infrared	30	
	Thermal	90	
ALOS	Fine beam: dual	12.5	May 17, 2006
PALSAR	Fine beam: single	12.5	October 2, 2006
	Fine beam: single	12.5	February 12, 2007

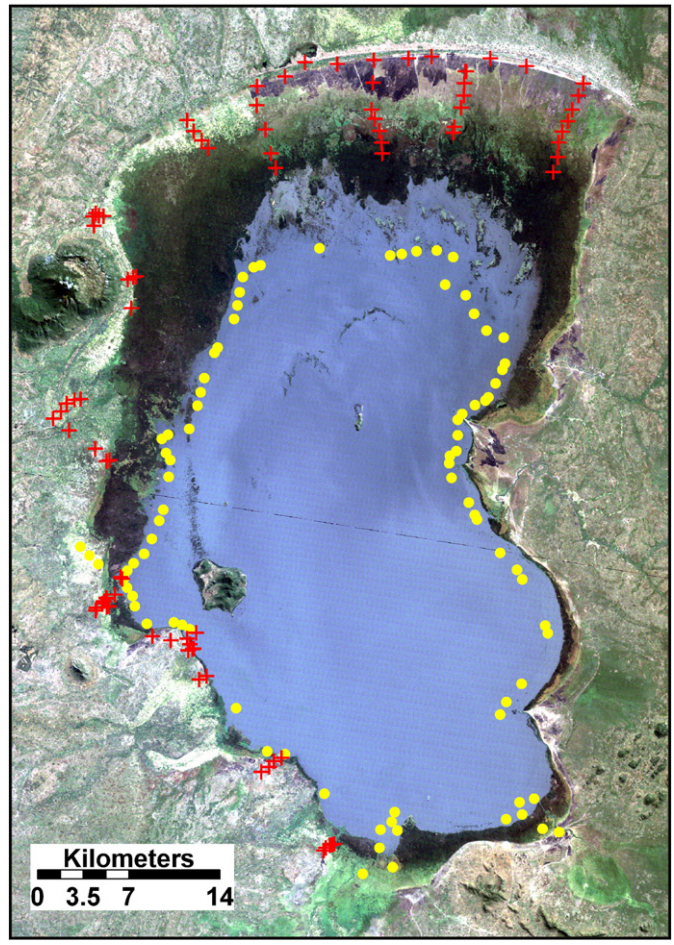

Fig. 4. Locations of field sites (crosses) and aerial photos (circles). Base image: Landsat TM real colour composite, November 2005.

Lake Chilwa fisheries

Lake Chilwa is one of the main fishing waters of Malawi, but fish production from the lake is highly variable. Fluctuations in Malawi's annual national fish catch are thus partly explained by the variations in the annual production from the Lake Chilwa ecosystem (Njaya, 2001). Data on the annual fish catch for Malawi have been extracted from the FishStat Plus database provided by the Food and Agriculture Organization (FAO) of the United Nations, for a 40-year time period. Total available catch data for each species has been aggregated to provide total fisheries production in tonnes per year at the country level, for the period 1992–2002. In addition, total catch data for Lake Chilwa for the same period have been acquired from the Fisheries Department in Zomba, Malawi. The data were analyzed to determine the Chilwa contribution in each year and the link between fisheries productivity and lake water level. It should be noted, however, that in the capture fisheries of Malawi, over 85% of annual production is

thought to come from the traditional sector, which may not be accurately documented, and therefore, the annual production figures are likely to be underestimates.

Results and discussion

Wetland hydrology, vegetation, and ecology

Table 4 presents a summary of the estimated mean annual water fluxes for the wetland. The estimate indicates that approximately 60% of the water in the wetland originates as direct rainfall onto the wetland, with 40% originating as flow from the surrounding catchment. The estimated average annual inflow (comprising the sum of surface and any groundwater inflows) of 1472 Mm^3 equates to the equivalent of an average annual contribution from the catchment (6039 km^2) of 244 mm. Assuming an average annual rainfall of over the catchment of 1362 mm (the arithmetic mean of several rain gauges located in the catchment), this corresponds to a 18% of the average annual rainfall.

They are based on many assumptions (described above), so the water budget results must be treated with caution. Without much more detailed hydrometric monitoring, there is no comprehensive way to verify the estimated water fluxes or to quantify the uncertainty. However, the total catchment area to the flow gauges on the four gauged rivers that flow into the lake is 1362 km^2 (i.e., 22.5% of the total catchment area to the lake). River discharge was measured at these stations over different periods of time, but the sum of the mean annual flow measured at these stations is 259 Mm^3 (i.e., 17.6% of the estimated total inflow derived from the water budget). Hence, if the gauged portion of the catchment is representative of the whole catchment, the measured flows suggest that the total inflow derived from the water budget calculation is approximately correct. The fact that it is slightly greater than would be estimated from the gauged flows alone may indicate that there is indeed a relatively small groundwater flow into the lake. Since the inflow estimate appears to be reasonable it is possible that the other fluxes estimated for the water budget are also more or less correct.

The Lake Chilwa ecosystem exhibits clear wetland zonation (Fig. 5). The site consists of three distinct components: open water (828 km^2), seasonal open water (303 km^2), and seasonally inundated marsh/ grassland floodplain (688 km^2). Open water refers to the area of the lake, which is inundated in both the wet and dry season of the study; seasonal open water refers to an area that is flooded in the wet season but, during lower lake levels during the dry season, consists of mud and macrophytes; and floodplain refers to the surrounding belt of *Typha* and grassland.

The northern section of the lake is bounded by a wide, dense belt of *Typha domingensis*. This is bordered by an area dominated by grasses (*Diplachne fusca* and *Panicum repens*), which gives way to a seasonally inundated grassland floodplain (with *Hyparrhenia rufa*, *Cynodon dactylon*, and *Sporobolus pyrramidalis*). The western edge of the lake supports *Vossia cuspidata* along with *Phragmites maurittianus* and *Cyperus papyrus*). The southern section of the lake is also bounded by a large emergent wetland plant (*T. domingensis*), although this area is much narrower than in the north and is surrounded by a transition zone, dominated by grass (*V. cuspidata*) and sedges (*Cyperus longus*) and interspersed with a shrub (*Aeschynomene phundii*), between the wetter and drier areas. The open water areas contain a number of submerged and floating plants (e.g., *Ceratophylum demersum*, *Pistia stratiotes*, *Utricularia reflexa*, *Nymphaea caerulea*, *Ipomoea aquatic*, and *Azolla* sp.). Although the entire Chilwa lake–wetland system undergoes significant changes in response to lake-level fluctuations, the marginal vegetation varies seasonally with water level variations, while the distribution of *T. domingensis* is more static (Kalk et al., 1979; Osborne, 2000).

The major land uses on the floodplain are livestock grazing, with much of the area overgrazed during the dry season, and cultivation of rice and other crops, including vegetables during the wet season. The *Typha* marsh is cut and burned extensively during the dry season, mainly to increase accessibility to the open water. Burning of rooted emergent vegetation such as this constitutes a major loss of organic matter and energy fixed by primary producers from the system (Osborne, 2000).

Table 4
Estimate of the average annual water fluxes (Mm^3) into and out of the Lake Chilwa wetland.

	Wet season	Dry season	Annual
Direct rainfall onto the wetland	2033	277	2310
Evapotranspiration from the wetland	2002	1780	3782
Outflow from the wetland	0	0	0
Inflow	1177	294	1472

Fire activity within the Lake Chilwa catchment

Monthly active fire detections within the basin are summarised in Table 5. A total of 1896 1-km × 1-km pixels were detected as containing a fire within the Chilwa basin during 2006. The majority of detections occurred during the months of July to October, with peak fire activity in August with 36% of all fires detected occurring during this month. Due to the different overpass times of the two satellites, it is possible to derive information related to the diurnal nature of the fire activity; for each month, more fires are detected within the basin during the afternoon overpass, with 76% of all detected fires occurring in the afternoon. This diurnal pattern of burning is typically observed in southern Africa as the ground and vegetation are drier and winds are stronger during the afternoon increasing the chances of ignition and encouraging fires to spread further (Rebelo, 2006).

The spatial location of fires has been mapped for the Chilwa basin along with the elevation data derived from the SRTM data set, with the areas below 625 masl corresponding to the location of Lake Chilwa (Fig. 6). It is clear from these data that there is a large spatial disparity in the locations of fires, with a larger number occurring in the Mozambique section of the basin in comparison to the Malawian side. These data are summarized in Table 6.

During 2006, despite the smaller area on the Mozambique side of the border (i.e., 35% of the catchment), 63% of all fires detected were located in the Mozambique side of the basin. The higher fire activity within the Mozambique section is therefore not due to greater land surface area. Within the Chilwa basin, active fires were detected on 40% of the land surface in Mozambique and on only 12% of the land surface of Malawi during 2006.

The Lake Chilwa fisheries

Africa's shallow lakes such as Lake Chilwa are among the most productive but also the most variable fishery ecosystems in the tropics (Allison and Mvula, 2002). Although the Lake Chilwa ecosystem is one of the most productive in Africa and is comparable to extensive aquaculture systems, it may still be susceptible to adverse ecological change (Jamu et al., 2006). While in 1979 and 1990, catches from Lake Chilwa constituted 43% and 31% of Malawi's total annual catches, respectively (Fig. 7), on average, between 1962 and 2003, this figure is much lower (16%), and during drying phases of the lake such as the 1995–1997 rainfall recession, fishing ceases completely. During productive years the annual catch is over 20,000 tonnes (25,800 tonnes in 1979 and 23,500 tonnes in 1990), but the average over the period of analysis (1962–2003) is approximately 9000 tonnes. During peak years, the productivity of Lake Chilwa is over four times greater than that of Lake Malawi. In 1990 for example, the productivity of the Lake Chilwa fisheries is 113 kg/ha (surface area of Lake Chilwa and surrounding

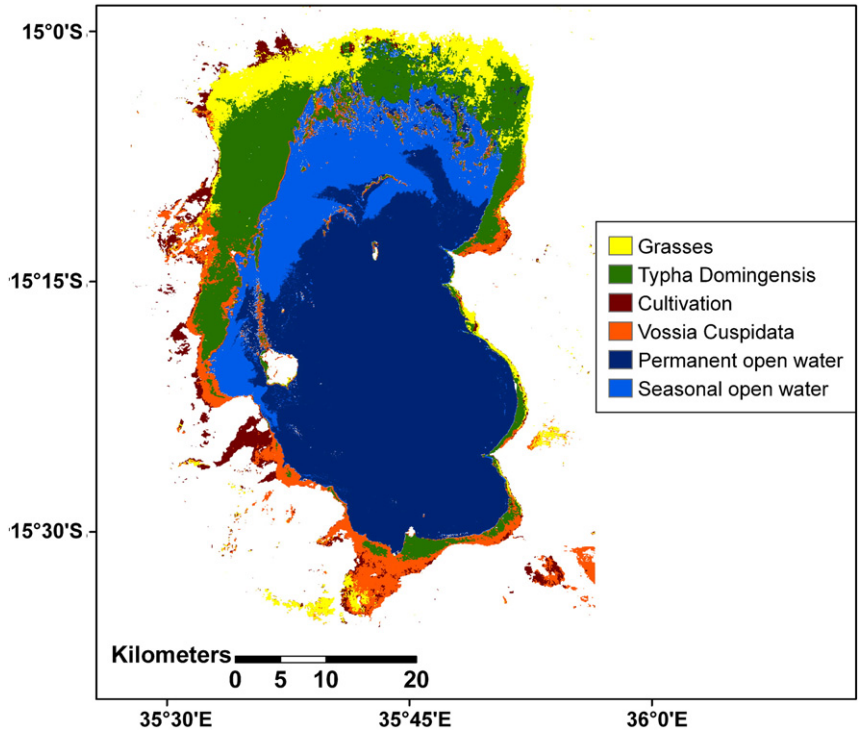

Fig. 5. Remote sensing derived land cover map for Lake Chilwa and Chiuta, 2006.

wetlands for 1990 is taken from Van Zegeren (1998)), while for Lake Malawi, the productivity in 1990 was approximately 25 kg/ha (surface area taken from the Lakenet Database: www.worldlakes.org).

The Lake Chilwa fisheries are clearly highly variable and are determined by the seasonal and long-term fluctuations in lake levels. The lake has experience recessions of varying degrees several times since 1879, with records indicating reduced catches and fish mortalities during the minor and larger recessions, and fisheries typically requiring 3 to 4 years to fully recover following a severe recession (Kalk et al., 1979; Njaya, 2001). Highest mean annual water levels were measured in 1990, and correspond to the second highest catch record of 23,500 tonnes, while the lowest water level recorded during the severe recession in 1996 corresponds to the lowest fish catch (0 tonnes). However, although Furse et al. (1979) state that yields and species composition in the lake vary as a function of lake level, correlation between annual catch and lake level is not clear on an interannual basis ($r^2 = 0.4$). It should also be noted that the gauging system of the lake was changed in 1989, and therefore, lake-level measurements from 1989 onwards may not be directly comparable to previous years (Allison and Mvula, 2002).

Table 5
Monthly fire activity within the Lake Chilwa catchment, 2006: number of fire pixels detected.

Month	MODIS Terra (morning overpass)	MODIS Aqua (afternoon overpass)	Total
May	5	58	63
June	24	52	76
July	39	139	178
August	127	559	686
September	138	370	508
October	112	223	335
November	13	28	41
December	0	9	9
Total	458	1438	1896

Resilience of the Lake Chilwa ecosystem

It is expected that the resilience of the Lake Chilwa ecosystem will be largely influenced by changes in the hydrology as a consequence of changes in the inflows. The climate change projections for the region indicate a decrease in available water resources with adverse impacts on agriculture (IPCC, 2007). At the same time, the lake ecosystem is under increasing pressure from changes in agriculture and fisheries as the human population rises and the demand for food increases. The "integrated small scale economy of farming, fishing and cattle-rearing" of the 1960s described by Kalk et al. (1979, p. 15) has changed to one where livestock rearing has decreased and cultivation (including irrigation) and fishing increased, potentially decreasing the resilience of the overall livelihood system (Allison and Mvula, 2002). More pressure on the Lake Chilwa fisheries is anticipated as the population increases and if fish products are traded with other communities.

Many of the recent changes in the watershed (e.g., siltation, damming, pollution, and land management) responsible for incidents of spawning failure are related to increased population pressure and poverty, which has resulted in the unsustainable use of water, forestry, and fishery resources (Jamu et al., 2006). Wood fuel (firewood and charcoal) is the dominant source of energy in Malawi's economy. It is a major source of energy for rural households and informal industries, such as fish smoking and processing (MNREA, 2002). Charcoal production is a major livelihood activity in the Lake Chilwa catchment. Undertaken in an unsustainable manner, charcoal production results in deforestation and land degradation, which will contribute to increased soil erosion and siltation of the lake.

In addition, as Lake Chilwa is located in a closed basin, accumulation of fertilizers, pesticides, and toxic chemicals may have a large influence on the lake; data on the extent of usage of chemicals are not readily available. As fisheries are highly prone to natural variability in their environment, the pressure from climate change is also expected to exacerbate the situation. The lake is shallow with high levels of evaporation, resulting in vulnerability to drying during prolonged dry spells. Moss (1979) highlights that while the Lake

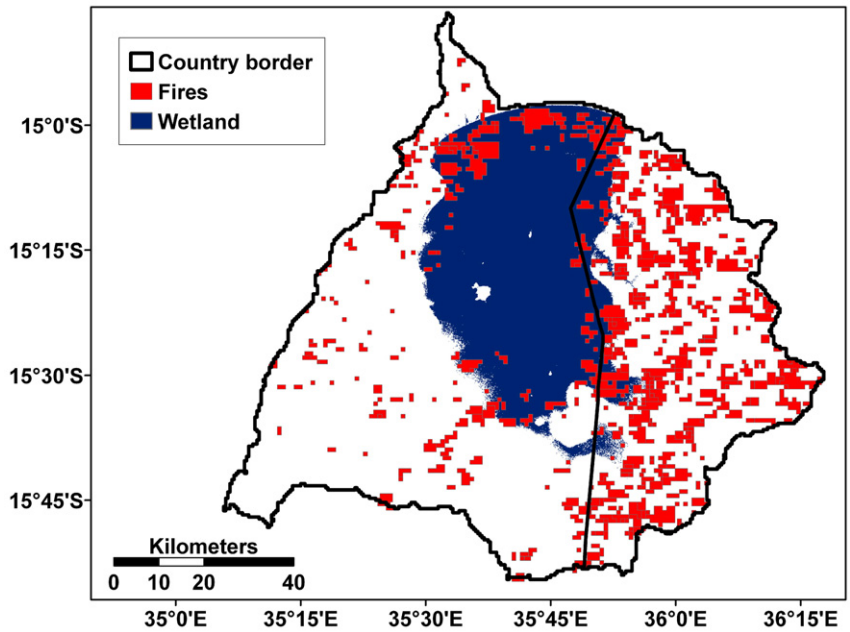

Fig. 6. Spatial locations of MODIS (Terra and Aqua) active fire detections across the Chilwa basin during 2006.

Chilwa fisheries are expected to persist in the unpredictable and variable Chilwa ecosystem, their resilience is dependent upon the maintenance of the refugium of marsh and streams that characterize the entire wetland ecosystem and cautions that threats to the marsh through reclamation for agriculture and siltation due to changes in land use within the catchment may have a greater impact on the system than overfishing.

Although most of the Lake Chilwa fish species are tolerant to a wide range of environmental factors and have the ability to adapt quickly to a changing environment, they are dependent on the ecological condition of the wetlands. Some of the most commercially valuable fish species found in Lake Chilwa including a small minow (*Barbus paludinosus*) and the African catfish (*Clarias garepinus*) make a lateral migration to the inundated floodplain and marsh during the rainy season to breed, returning to the open lake soon afterwards (Njaya, 2001). Future reduction in lake levels due to both increased extraction in the catchment particularly for irrigation, and climatic variability, may result in failure of the lake to inundate the extensive marsh and grassland floodplain seasonally, thereby greatly reducing the productivity of the lake and, in particular, the fisheries (Jamu et al., 2006). Sarch and Allison (2000) note the greater importance of climate relative to fishing in driving dynamics of fish stocks in African inland lakes such as Lake Chilwa and suggest that efforts should therefore be redirected to protecting wetland functions and broader ecosystem integrity and away from trying to manage fish stocks for sustainability. The importance of peripheral floodplains and marsh and marginal areas of shallow lakes has been emphasized, leading to the recommendation that management policies for shallow lakes should extend as much as possible to the surrounding land (Kalk et al, 1979; *UNEP*, 1981; Njaya, 2001). For the Lake Chilwa ecosystem, this involves the maintenance of the areas of *Typha* that surround a large part of the lake, and are subject to extensive burning.

Table 6
Fire activity within the Lake Chilwa catchment, 2006.

Country	MODIS Terra (morning overpass)	MODIS Aqua (afternoon overpass)	Total
Malawi	195	505	700
Mozambique	263	933	1196
Total	458	1438	1896

Although a management plan exists for the lake, this currently does not take into account transboundary issues. Due to differences in environmental and land use practices between the third of the catchment area that falls in Mozambique and the Malawi portion, a transboundary approach to management is important. The Mozambican side has a much lower population density (17 people/km^2 compared to 162 people/km^2 in Malawi) and has relatively undisturbed forests and a low level of agricultural activity, in comparison to intensive and unsustainable land use practices, deforestation, and high levels of water extraction in the Malawi component (Jamu et al., 2006). Fire activity is much higher in Mozambique. This is likely due to the wider distribution of relatively undisturbed forests. The "natural" vegetation consists predominantly of Miombo woodlands and Combretum-Acacia savanna; frequent fires are a characteristic of these fire climax vegetation types. In addition, due to lower population densities and less fragmented land cover, fires are likely to spread further than in the more fragmented landscape of the Malawian side of the catchment.

Despite numerous studies conducted into the Lake Chilwa ecosystem, none has examined the practice and frequency of burning within the catchment or specifically within the wetland. In a study looking at Indigenous Knowledge within the basin and perceived causes of lake resource degradation, 85% of respondents reported a drastic decrease in *T. domingensis* in recent years, also noting that this destruction might be responsible for the declining catches of fish (Kalanda-Sabola, 2007). The situation was believed to have worsened since 1995 as a result of extensive burning to clear the land for agriculture, with continuous fires restricting the regeneration of the *Typha*. Although very little research has examined the effect of frequent burning on the hydrology or water quality within African ecosystems, frequent fires may increase sediment and nutrient fluxes into the lake, and the frequent deposition of charcoal and ash may have an effect on the water quality. Although the effects of fire on hydrology will depend on the vegetation, fire severity, and climatic events after the fire (Doerr and Shakesby, 2006), erosion, sedimentation, and run-off have been shown to be significantly higher in a burned catchment than a comparable unburned catchment in a dryland area (Mayor et al., 2007). In particular, due to the closed nature of the Chilwa Basin and the shallowness of the lake, frequent

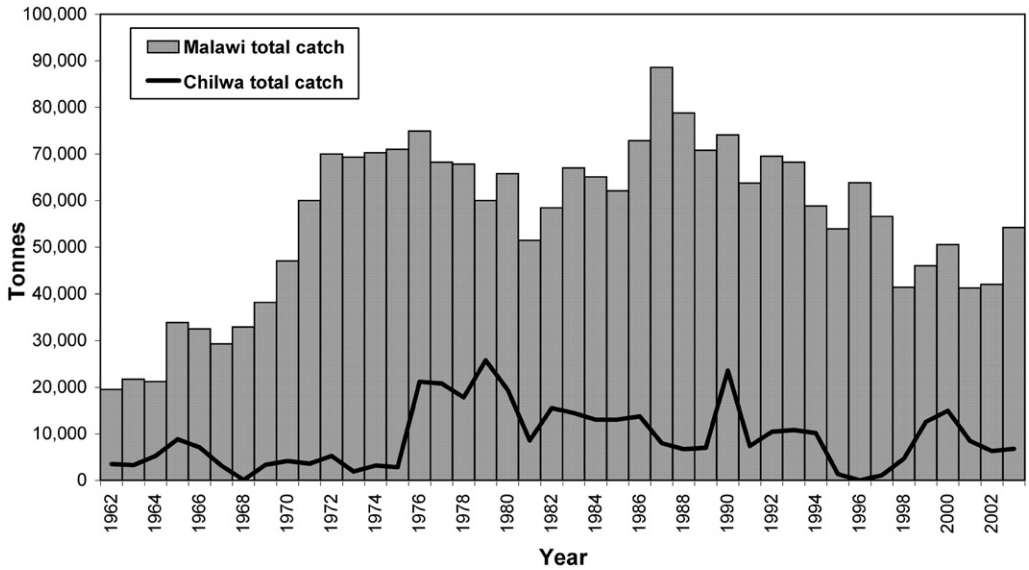

Fig. 7. Total annual catch statistics for Malawi and Lake Chilwa (FAOSTAT, 2008; Fisheries Department, 2008).

burning may have a negative impact on the resilience of the ecosystem. As fire occurrence, intensity, and severity are likely to increase under future climate change scenarios, it would be prudent for sustainable land management approaches detailed within the wetland management plan to incorporate a strategy on burning practices within the catchment.

Lake Chilwa is an extremely productive lake and has been nominated by Malawi as a wetland of International Importance. It is, however, a highly variable ecosystem and subject to increasing pressure that may undermine its ability to respond to changes in either the climate/hydrology or nutrient balance and harvesting of fish stocks. While the wetlands are critical habitats providing resilience to the lake, the sustainable management of land use practices within the catchment is essential to the long-term resilience of the entire Lake Chilwa ecosystem. As the lake and catchment is shared with Mozambique, a common strategy is needed for the two countries. Given the pressures on the lake, there is clearly a need for an integrated, adaptive, and transboundary management plan that encompasses both the Malawian and Mozambican components of the catchment and which addresses not only the hydrology and lake fisheries but also the land use practices within the wetland and the surrounding catchment, including the extent of irrigation and other cropping and grazing practices.

Conclusions

Malawi has a large network of lakes, freshwater marshes, and floodplains. These areas support millions of livelihoods and greatly contribute to the food security of the country. However, very little baseline data exists for these ecosystems. Characterisation of Malawi's wetlands is problematic because of their inaccessible nature. This study has illustrated how remote sensing data can be used along with ground surveys to provide a valuable spatial characterization of a wetland. The maps of the Lake Chilwa wetland and its catchment thus produced, used in conjunction with information on land use and livelihood activities, provide invaluable information with which to assess threats to the wetlands ecological character and its vulnerability. Baseline information, such as that described in this study on the distribution, ecology, hydrology, and land use of wetlands as well as their surrounding catchments, is essential to identify limits to resilience and hence is a prerequisite for developing sustainable management plans. For Lake Chilwa, the data presented highlight the need for a common catchment management strategy between Malawi and Mozambique, as well as the need for an adaptive approach to management of the wetland. This is of particular importance due to the seasonal and periodic water-level fluctuations and frequent drying up of the lake, which result in shifts in livelihood activities within the wetland and the catchment. The need to enhance the flexibility of livelihoods is therefore also crucial. Currently, Malawi has no policy for the wise use of its wetlands. The resilience of Malawi's wetlands and in particular the Lake Chilwa ecosystem under likely future climate change, increasing population, and increasing pressure on resources will depend on their sustainable use. It is therefore essential that mechanisms are put in place to minimize degradation of Malawi's lakes and wetlands, to optimize the ecosystem and livelihood benefits.

Acknowledgments

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